Table B1. Spline-indicator fixed-parameter binary logit model of crash injury severity (serious injury/fatal vs. PDO/minor injury), single-motorcycle crashes, Thailand, 2016–2023: implied parameter for each two-year period, estimated parameter (*t*-statistic).

Variable description	Spline terms retained (*j*)	2016–2017	2018–2019	2020–2021	2022–2023
Constant [serious injury / fatal]					
Constant	1, 3	0.183 (2.65)	0.366 (2.65)	0.055 (0.36)	–0.256 (–1.19)
Vehicle involvement					
Pedestrian involved (1 if yes, 0 otherwise)	1	–0.037 (–0.67)	–0.075 (–0.67)	–0.112 (–0.67)	–0.149 (–0.67)
Crash type					
Rear-end collision in the same lane (1 if yes, 0 otherwise)	1, 3	–0.123 (–1.22)	–0.246 (–1.22)	0.248 (1.42)	0.743 (2.47)
Ran off road to the left and hit an object (1 if yes, 0 otherwise)	1	0.138 (2.92)	0.277 (2.92)	0.415 (2.92)	0.554 (2.92)
Vehicle mounted a traffic island (1 if yes, 0 otherwise)	1	0.039 (0.77)	0.078 (0.77)	0.117 (0.77)	0.156 (0.77)
Sideswipe (parallel movement) (1 if yes, 0 otherwise)	1, 3, 4	–0.216 (–1.76)	–0.432 (–1.76)	0.692 (2.05)	–0.070 (–0.23)
Ran off road to the left (1 if yes, 0 otherwise)	1, 2, 3	–0.148 (–1.26)	–0.818 (–6.61)	–0.480 (–4.55)	–0.142 (–0.88)
Ran off road to the right (1 if yes, 0 otherwise)	1, 3	–0.291 (–3.31)	–0.582 (–3.31)	–0.381 (–2.75)	–0.181 (–0.82)
Loss of vehicle control (1 if yes, 0 otherwise)	1, 2, 3	–0.042 (–0.23)	–1.238 (–8.20)	–1.066 (–9.13)	–0.893 (–5.00)
Ran off road on a left bend (1 if yes, 0 otherwise)	1, 4	–0.126 (–1.91)	–0.252 (–1.91)	–0.378 (–1.91)	0.186 (0.65)
Ran off road on a right bend (1 if yes, 0 otherwise)	1, 2, 3	0.315 (1.54)	–0.613 (–3.10)	–0.246 (–1.57)	0.121 (0.49)
Crash cause					
Speeding (1 if yes, 0 otherwise)	1	0.084 (2.14)	0.167 (2.14)	0.251 (2.14)	0.335 (2.14)
Cutting in too closely (1 if yes, 0 otherwise)	1, 3, 4	0.156 (1.57)	0.313 (1.57)	–0.607 (–2.21)	–0.043 (–0.14)
Driver fell asleep (1 if yes, 0 otherwise)	1, 4	0.102 (0.95)	0.204 (0.95)	0.306 (0.95)	1.336 (3.74)
Drunk driving (1 if yes, 0 otherwise)	1, 2, 4	–0.297 (–2.12)	0.040 (0.33)	0.378 (1.84)	0.137 (0.53)
Roadway characteristics					
Crash on the main carriageway (1 if yes, 0 otherwise)	1, 2	–0.450 (–2.31)	–0.083 (–0.61)	0.284 (1.86)	0.651 (2.84)
Crash on a frontage road (1 if yes, 0 otherwise)	1, 2	–1.086 (–4.73)	–0.719 (–4.70)	–0.352 (–2.40)	0.015 (0.07)
Road under construction (1 if yes, 0 otherwise)	1, 3	0.347 (2.25)	0.693 (2.25)	0.287 (1.70)	–0.119 (–0.61)
Painted median (1 if yes, 0 otherwise)	1, 3	0.114 (1.49)	0.227 (1.49)	–0.166 (–1.43)	–0.560 (–2.81)
Raised or earth median (1 if yes, 0 otherwise)	1, 2, 3	–0.400 (–3.37)	–0.020 (–0.17)	–0.071 (–0.85)	–0.123 (–0.96)
Depressed median (1 if yes, 0 otherwise)	1	0.087 (2.92)	0.173 (2.92)	0.260 (2.92)	0.347 (2.92)
Barrier median (1 if yes, 0 otherwise)	1, 2, 3	–0.595 (–3.17)	0.058 (0.32)	–0.265 (–2.27)	–0.588 (–3.49)
Horizontal curve (1 if yes, 0 otherwise)	1	0.032 (0.74)	0.064 (0.74)	0.095 (0.74)	0.127 (0.74)
Concrete pavement surface (1 if yes, 0 otherwise)	1	–0.090 (–3.14)	–0.181 (–3.14)	–0.271 (–3.14)	–0.361 (–3.14)
T-intersection (1 if yes, 0 otherwise)	1, 4	–0.163 (–1.72)	–0.325 (–1.72)	–0.488 (–1.72)	0.158 (0.59)
Environmental characteristics					
Wet road surface (1 if yes, 0 otherwise)	1	0.023 (0.23)	0.045 (0.23)	0.068 (0.23)	0.090 (0.23)
Raining (1 if yes, 0 otherwise)	1, 3, 4	0.305 (2.03)	0.611 (2.03)	–0.500 (–1.38)	0.109 (0.24)
Dark, no street lighting (1 if yes, 0 otherwise)	1, 2	1.298 (4.92)	0.914 (5.20)	0.529 (2.85)	0.145 (0.51)
Dark, with street lighting (1 if yes, 0 otherwise)	1, 2	1.027 (4.17)	0.739 (4.49)	0.452 (2.59)	0.164 (0.62)
Temporal characteristics					
Weekend (1 if yes, 0 otherwise)	1	–0.013 (–0.63)	–0.025 (–0.63)	–0.038 (–0.63)	–0.051 (–0.63)
Songkran festival period (1 if yes, 0 otherwise)	1, 2	–0.509 (–4.93)	–0.545 (–7.69)	–0.581 (–7.29)	–0.617 (–5.10)
New Year festival period (1 if yes, 0 otherwise)	1, 3	–0.409 (–8.03)	–0.817 (–8.03)	–0.635 (–8.61)	–0.454 (–3.67)
Evening (1 if yes, 0 otherwise)	1, 2	–0.847 (–3.33)	–0.550 (–3.18)	–0.254 (–1.36)	0.043 (0.15)
Midnight / early morning (1 if yes, 0 otherwise)	1, 2	–0.305 (–1.27)	0.009 (0.05)	0.323 (1.83)	0.637 (2.37)
Peak hours (1 if yes, 0 otherwise)	1	–0.038 (–1.26)	–0.076 (–1.26)	–0.114 (–1.26)	–0.152 (–1.26)
Region					
Eastern region (1 if yes, 0 otherwise)	1, 2	0.840 (4.63)	0.651 (5.35)	0.462 (3.94)	0.273 (1.58)
Northeastern region (1 if yes, 0 otherwise)	1, 2	0.442 (4.15)	0.579 (7.73)	0.717 (8.26)	0.854 (6.54)
Northern region (1 if yes, 0 otherwise)	1	–0.061 (–2.14)	–0.122 (–2.14)	–0.184 (–2.14)	–0.245 (–2.14)
Southern region (1 if yes, 0 otherwise)	1	0.006 (0.20)	0.013 (0.20)	0.019 (0.20)	0.025 (0.20)

Variable description	Spline terms retained (j)	2016–2017	2018–2019	2020–2021	2022–2023
Western region (1 if yes, 0 otherwise)	1, 3, 4	0.013 (0.16)	0.026 (0.16)	1.010 (4.12)	0.186 (0.94)
Model statistics					
Number of observations			7,325		
Number of estimated spline coefficients			78		
Variables with the base term only (parameter proportional to the period index)			12 of 40		
Log-likelihood at zero, LL(0)			–5077.30		
Log-likelihood with constant only, LL(C)			–5074.03		
Log-likelihood at convergence, LL(β)			–4558.15		
McFadden pseudo $\rho^2 = 1 - LL(\beta)/LL(0)$			0.102		
AIC / BIC			9,272.3 / 9,810.4		
LR test against the fixed-parameter separate-period models			$\chi^2 = 74.46$, df = 82, confidence of rejection = 0.289		

Notes. A positive parameter increases the probability of a serious injury/fatal outcome. Reference categories: Central region, daytime, no median, one motorcycle. Following Shabab et al. (2024), each variable enters through spline terms $f_j(\tau) = \max(\tau - j + 1, 0)$, $j = 1, \ldots, 4$, where $\tau = 1, \ldots, 4$ indexes the two-year periods; the parameter in two-year period τ is $\beta(\tau) = \sum_j b_j f_j(\tau)$. The base term ($j = 1$) is always retained; the other terms were removed by backward elimination at the 90% level ($|t| < 1.645$, one term per variable per round). A variable with the base term only has a parameter proportional to the period index. The table reports the implied parameters $\beta(\tau)$; t-statistics by the delta method from the covariance of the spline coefficients.